Everolimus Monotherapy

INDICATIONS FOR USE:

INDICATION	ICD10	Regimen Code	HSE approved reimbursement status*
Treatment of patients with advanced renal cell carcinoma, whose disease has progressed on or after treatment with VEGF-targeted therapy.	C64	00320a	N/A
Treatment of unresectable or metastatic, well- or moderately-differentiated neuroendocrine tumours of pancreatic origin in adults with progressive disease.	C25	00320b	N/A
The treatment of unresectable or metastatic, well-differentiated (Grade 1 or Grade 2) non-functional neuroendocrine tumours of gastrointestinal origin in adults with progressive disease.	C25	00320c	N/A

* This applies to post 2012 indications

TREATMENT:

The starting dose of the drugs detailed below may be adjusted downward by the prescribing clinician, using their independent medical judgement, to consider each patients individual clinical circumstances.

Drug	Dose	Route	Cycle
Everolimus	10mg once daily	PO	Continuous
Tablet to be taken at the same time every day, consistently either with or without food. Swallow whole with a glass of water			
If a dose is missed, the patient should not take an additional dose, but take the next prescribed dose as usual.			
The tablets should **not** be chewed or crushed.			
Everolimus is commonly available as 2.5mg/5mg and 10mg tablets.			

ELIGIBILITY:

- Indications as above
- ECOG status 0-2

EXCLUSIONS:

- Hypersensitivity to everolimus, to other rapamycin derivatives or any of the excipients
- Caution is advised for patients with pre-existing significant lung compromise due to the risk for pneumonitis
- Pregnancy
- Breastfeeding

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PRESCRIPTIVE AUTHORITY:

The treatment plan must be initiated by a Consultant Medical Oncologist

TESTS:

Baseline tests:

- FBC, renal and liver profile
- Blood urea nitrogen (BUN), urinary protein
- Total cholesterol and triglycerides
- Fasting serum glucose
- Virology screen - Hepatitis B (HBsAg, HBcoreAb)
  *(Reference Regimen Specific Complications for information on Hepatitis B reactivation)

Regular tests:

- FBC, renal and liver profile every 4 weeks
- Total cholesterol and triglycerides
- Monitoring of fasting serum glucose as clinically indicated

Disease monitoring:

Disease monitoring should be in line with the patient's treatment plan and any other test/s as directed by the supervising Consultant.

DOSE MODIFICATIONS:

- Any dose modification should be discussed with a Consultant
- Management of severe and/or intolerable suspected adverse reactions may require dose reduction and/or temporary interruption of everolimus therapy
- For adverse reactions of Grade 1, dose adjustment is usually not required

Haematological:

Table 1: Dose modification of everolimus in haematological toxicity

ANC ($x10^9$/L)		Platelets ($x10^9$/L)	Dose recommendation
≥1		≥ 75	No dose adjustment required.
0.5-0.99	and/or	50-74.9	Temporary dose interruption until neutrophils recover to ≥1 x 10^9/L and/or platelets recover to ≥75x10^9/l. Re-initiate treatment at same dose.
<0.5	and/or	<50	Temporary dose interruption until neutrophils recover to ≥1 x 10^9/L and/or platelets recover to ≥75x10^9/l. Re-initiate treatment at 5mg daily
Grade 3 Febrile Neutropenia			Temporary dose interruption until recovery to Grade 2 (≥1.25x10^9/L) and no fever. Re-initiate treatment at 5 mg daily.
Grade 4 Febrile Neutropenia			Discontinue treatment

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Table 2: Dose modification of everolimus in renal and hepatic impairment

Renal Impairment	Hepatic Impairment*	
No dose adjustment required	Child-Pugh Class A	75% of the original dose
	Child-Pugh Class B	50% of the original dose
	Child-Pugh Class C	25% of the original dose
Recommendations for Everolimus from Giraud et al 2023		

*Dose adjustments should be made if a patient's hepatic (Child-Pugh) status changes during treatment

Management of adverse events:

Table 3: Dose Modification of everolimus for Adverse Events

Adverse reactions	Recommended dose modification
Non-infectious pneumonitis	
• Grade 2	Consider interruption of therapy until symptoms improve to Grade ≤ 1. Re-initiate treatment at 5 mg daily. Discontinue treatment if failure to recover within 4 weeks.
• Grade 3	Interrupt treatment until symptoms resolve to Grade ≤ 1. Consider re-initiating treatment at 5mg daily. If toxicity recurs at Grade 3, consider discontinuation.
• Grade 4	Discontinue treatment.
Stomatitis	
• Grade 2	
○ 1st occurrence	Temporary dose interruption until recovery to Grade ≤ 1. Re-initiate treatment at same dose.
○ 2nd occurrence	Interrupt dose until recovery to Grade ≤ 1. Re-initiate treatment at 5 mg daily.
• Grade 3	Temporary dose interruption until recovery to Grade ≤ 1. Re-initiate treatment at 5 mg daily.
• Grade 4	Discontinue treatment
Other non-haematological toxicities (excluding metabolic events)	
• Grade 2	
○ 1st occurrence	If toxicity is tolerable, no dose adjustment required. If toxicity becomes intolerable, temporary dose interruption until recovery to Grade ≤1. Re-initiate treatment at **same** dose.
○ 2nd occurrence	If toxicity recurs at Grade 2, interrupt treatment until recovery to Grade ≤1. Re-initiate treatment at 5 mg daily.
• Grade 3	
○ 1st occurrence	Temporary dose interruption until recovery to Grade ≤1. Consider re-initiating treatment at 5 mg daily.
○ 2nd occurrence	Consider discontinuation.
• Grade 4	Discontinue treatment.
Metabolic events (e.g. hyperglycaemia, dyslipidaemia)	
• Grade 2	No dose adjustment required.
• Grade 3	Temporary dose interruption. Re-initiate treatment at 5 mg daily.
• Grade 4	Discontinue treatment.

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SUPPORTIVE CARE:

EMETOGENIC POTENTIAL:

- As outlined in NCCP Classification Document for Systemic AntiCancer Therapy (SACT) Induced Nausea and Vomiting Available on the NCCP website

Everolimus: Minimal to Low (**Refer to local policy)**

For information:
Within NCIS regimens, antiemetics have been standardised by the Medical Oncologists and Haemato-oncologists and information is available in the following documents:

- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Medical Oncology) - Available on the NCCP website
- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Haemato-oncology) - Available on the NCCP website

PREMEDICATIONS: None required

OTHER SUPPORTIVE CARE:

- The use of non-alcoholic prophylactic or therapeutic mouthwashes may be required for the prevention or management of mucositis **(Refer to local policy)**
- Everolimus may have a minor or moderate influence on the ability to drive and use machines. Patients should be advised to be cautious when driving or using machines if they experience fatigue during treatment with everolimus

ADVERSE EFFECTS

- Please refer to the relevant Summary of Product Characteristics (SmPC) for details

REGIMEN SPECIFIC COMPLICATIONS

- Please refer to the relevant Summary of Product Characteristics (SmPC) for details
- **Non-infectious pneumonitis:** This is a class effect of rapamycin derivatives, including everolimus. A diagnosis of non-infectious pneumonitis should be considered in patients presenting with non-specific respiratory signs and symptoms such as hypoxia, pleural effusion, cough or dyspnoea, and in whom infectious, neoplastic and other non-medicinal causes have been excluded by means of appropriate investigations. Opportunistic infections such as pneumocystis jirovecii (carinii) pneumonia (PJP, PCP) should be ruled out in the differential diagnosis of non-infectious pneumonitis. Patients should be advised to report promptly any new or worsening respiratory symptoms
    - Patients who develop radiological changes suggestive of non-infectious pneumonitis and have few or no symptoms may continue everolimus therapy without dose adjustments
    - If symptoms are moderate (Grade 2) or severe (Grade 3) the use of corticosteroids may be indicated until clinical symptoms resolve

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  - For patients who require use of corticosteroids for treatment of non-infectious pneumonitis, prophylaxis for pneumocystis jirovecii (carinii) pneumonia (PJP, PCP) may be considered

- **Infections:** Everolimus has immunosuppressive properties and may predispose patients to bacterial, fungal, viral or protozoan infections, including infections with opportunistic pathogens. Physicians and patients should be aware of the increased risk of infection with everolimus. Pre-existing infections should be treated appropriately and should have resolved fully before starting treatment with everolimus. While taking everolimus, be vigilant for symptoms and signs of infection; if a diagnosis of infection is made, institute appropriate treatment promptly and consider interruption or discontinuation of everolimus
- **Hypersensitivity reactions:** Hypersensitivity reactions manifested by symptoms including, but not limited to, anaphylaxis, dyspnoea, flushing, chest pain or angioedema (e.g. swelling of the airways or tongue, with or without respiratory impairment) have been observed with everolimus
- **Oral mucositis:** This is a common side effect with this treatment and may manifest as mouth and tongue ulceration. Early intervention may help to avoid dose alteration or interruption. Topical treatments (alcohol free) are recommended
- **Wound healing complications:** Impaired wound healing is a class effect of rapamycin derivatives, including everolimus. Caution should therefore be exercised with the use of everolimus in the peri-surgical period
- **Renal failure events:** Cases of renal failure (including acute renal failure), some with a fatal outcome, have been observed in patients treated with everolimus. Renal function should be monitored particularly where patients have additional risk factors that may further impair renal function.
- **Hepatitis B Reactivation:** Patients should be tested for both HBsAg and HBcoreAb as per local policy If either test is positive, such patients should be treated with anti-viral therapy. **(Refer to local infectious disease policy).** These patients should be considered for assessment by hepatology

DRUG INTERACTIONS:

- Current SmPC and drug interaction databases should be consulted for information

REFERENCES:

1. Motzer RJ, Escudier B, Oudard S, et al. Efficacy of everolimus in advanced renal cell carcinoma: a double-blind, randomised, placebo-controlled phase III trial. Lancet 2008;372:449-56.
2. Motzer RJ, Escudier B, Oudard S, et al. Phase 3 trial of everolimus for metastatic renal cell carcinoma : final results and analysis of prognostic factors.Cancer 2010;116(18):4256-4265.
3. Yao, J et al. Everolimus for advanced pancreatic neuroendocrine tumors. N Engl J Med 2011;364;6:514-23.
4. Yao, J et al. Everolimus for the treatment of advanced, non-functional neuroendocrine tumours of the lung or gastrointestinal tract (RADIANT-4): a randomised, placebo-controlled, phase 3 study. Lancet, The, 2016-03-05, Volume 387, Issue 10022, Pages 968-977
5. NCCP Classification Document for Systemic Anti-Cancer Therapy (SACT) Induced Nausea and Vomiting. V6 2025. Available at: https://www.hse.ie/eng/services/list/5/cancer/profinfo/chemoprotocols/nccp-classification-document-for-systemic-anti-cancer-therapy-sact-induced-nausea-and-vomiting.pdf

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6. Everolimus (Afinitor®) Summary of Product Characteristics. Last updated 20/11/2025. Accessed December 2025. Available at: https://www.ema.europa.eu/en/documents/product-information/afinitor-epar-product-information_en.pdf

Version	Date	Amendment	Approved By
1	3/5/2016		Prof Maccon Keane
2	3/5/2018	Applied new NCCP regimen template and updated dosing for adverse events as per SmPC	Prof Maccon Keane
3	14/01/2019	Added in new indication 00320c Added in virology screening	Prof Maccon Keane
4	22/01/2021	Updated hepatitis B reactivation standard wording	Prof Maccon Keane
5	18/10/2023	Removal of viral screening from baseline tests	Prof Maccon Keane
6	24/02/2026	Reviewed. Renal and hepatic dose modifications table updated in line with Giraud et al 2023. Virology added to baseline tests. Regimen updated in line with NCCP standardisation.	Prof Maccon Keane

Comments and feedback welcome at oncologydrugs@cancercontrol.ie.

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